# A local characteristic precursor for the registered Einstein–scalar shell packet

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**Abstract**

We construct a short-past linear precursor of a registered compact Einstein–scalar shell perturbation. The actual master data have nonzero velocity. A fresh rational first-exit calculation verifies the static positive collar and bounds the master potential at negative times by $5500 < \mathcal{W} < 103000$. A backward Volterra construction then yields the coupled first perturbation on $-1 < x \le 0$, before shell contact. On every such Cauchy section, both normalized principal quadratic shares lie strictly between 0.499999 and 0.500001. On a characteristic half-diamond with interior past apex, the solution gives two compatible past traces. Each exceeds 0.499999 at its midpoint, whereas their common corner value is below $-4\,10^{-8}$; each therefore has an internal zero and is nonconstant. A second Volterra formula reconstructs the same solution from these traces. This is a local linear construction and a finite-interval obstruction to specified principal one-way data. It does not select an earlier physical incoming state, complete a nonlinear null/shell problem, or determine a second-order preparation.

## 1 The fixed coupled packet and the question

A compact packet supplied on a present Cauchy slice need not be a pure traveling channel there. This fact alone says little about a more remote incoming state: a potential or a shell can mix characteristic channels. We examine a smaller, quantitative question for the registered packet: what is its unique local linear past before shell contact, and what data does it induce on two intersecting past characteristics? The construction uses standard wave and Volterra methods. Finite-slab wave estimates and time reversal are established tools [1]; no new general energy or characteristic principle is asserted.

The retained five-dimensional bulk action and one-copy outward shell laws are

$$\begin{aligned} S_{\mathrm{bulk}} &= \int \left( \frac{R}{2} - \frac{|\nabla\phi|^2}{2} - V_{\mathrm{b}}(\phi) \right) \mathrm{d}\mu_g, \\ \mathrm{Ric}_{AB} &= \partial_A\phi\,\partial_B\phi + \frac{2}{3} V_{\mathrm{b}} g_{AB}, \qquad \Box_g\phi = V_{\mathrm{b}}', \quad K^{\mathrm{out}}_{\mu\nu} = \frac{\sigma}{6}\gamma_{\mu\nu}, \quad n_{\mathrm{out}}\phi = -\frac{\sigma'}{2}. \end{aligned} \tag{1}$$

Two identical interiors meet at a pure-tension shell. The fixed model has

$$\begin{aligned} S(\phi) &= 1 - \phi + \phi^3/3, \qquad V_{\mathrm{b}} = (S')^2/2 - 2S^2/3, \\ \sigma &= 2S + 10^{-3}(1 + c_\sigma\phi), \quad c_\sigma = \frac{96714321425692}{161839258930731}. \end{aligned} \tag{2}$$

The static spherical background, with unit three-sphere metric $\kappa$, is

$$g_0 = \mathrm{d}y^2 + \rho^2[-\mathrm{d}t^2 + \cosh^2 t\,\kappa], \qquad \mathrm{d}y = -\rho\,\mathrm{d}z, \qquad a = \rho_y/\rho, \quad p = \phi_{0,y}. \tag{3}$$

The shell is $z = 0$ and increasing $z$ points into the positive collar, away from it. Write $\rho_b = \rho(0)$, $p_b = p(0)$ and $\eta = \phi_0 - 1$. No measured SI scale is assigned to these model-normalized coordinates.

**Retained premise 1.** *The actual smooth static branch, its selected analytic shell enclosure, the all-point first master reconstruction, and its canonical and common-harmonic one-jet identification are retained analytic inputs. The fresh arithmetic below verifies a collar and potential enclosure from those inputs. Hash authentication does not reprove the analytic identities or an exact-family existence theorem.*

For the longitudinal first master $\mathsf{V}$, put

$$\begin{gathered} C = \rho(3a + 2p_y/p), \quad \alpha = (\rho/\rho_b)^{3/2} p/p_b > 0, \quad \alpha_z/\alpha = -C/2, \\ \mathsf{V} = \alpha \cosh^{-3/2} t \, w, \qquad w_{tt} - w_{zz} + \mathcal{W}(t,z) w = 0, \\ \mathcal{W} = \frac{35}{4} \operatorname{sech}^2 t - \frac{25}{4} + \frac{C_z}{2} + \frac{C^2}{4} + \frac{2}{3}\rho^2 p^2. \end{gathered} \tag{4}$$

The registered actual data are

$$\begin{gathered} w(0,z) = q_\ell(z), \qquad\qquad w_t(0,z) = e^{-z} q_\ell(z), \quad \ell = 10^{-6}, \\ q_\ell(z) = \begin{cases} \exp\left(4 - \dfrac{\ell^2}{(z-\ell)(2\ell - z)}\right), & \ell < z < 2\ell, \\ 0, & \text{otherwise.} \end{cases} \end{gathered} \tag{5}$$

They are smooth and flat at the support endpoints. Their first velocity is not zero. The native 31 October change concerned a second-response approximation and does not change this first state [3]. Set $x = t/\ell$, $r = z/\ell$, $U(x,r) = w(\ell x, \ell r)$, and $Q(r) = q_\ell(\ell r)$. Then

$$U_{xx} - U_{rr} + \mathcal{V} U = 0, \quad \mathcal{V} = \ell^2 \mathcal{W}(\ell x, \ell r), \quad U(0,r) = Q(r), \quad U_x(0,r) = \ell e^{-\ell r} Q(r). \tag{6}$$

We retain the future-oriented definitions $J_+ = U_x + U_r$ and $J_- = U_x - U_r$. Their principal speeds are $\mathrm{d}r/\,\mathrm{d}x = -1$ toward the shell and $+1$ away, respectively.

## 2 Fresh negative-time coefficients on the actual collar

The selected wide-collar input declares $0 \le z \le 8\ell$, with outer box

$$\begin{gathered} 129 \le \rho \le 131, \qquad -9/10^5 \le \eta \le -8/10^5, \\ -7/50000 \le p \le -3/25000, \qquad 111/1000 \le a \le 14/125. \end{gathered} \tag{7}$$

This width is checked afresh; no narrower historical collar is enlarged by assertion. Let $U_j = \partial_\eta^j V_{\mathrm{b}}(1+\eta)$. The static ODE is

$$\rho_z = -\rho^2 a, \qquad \eta_z = -\rho p, \qquad p_z = -\rho(U_1 - 4ap), \qquad a_z = \rho^{-1} + \rho p^2/3. \tag{8}$$

Using the selected shell intervals and the actual polynomial

$$V_{\mathrm{b}}(1+\eta) = -\frac{2}{27} + \frac{14}{9}\eta^2 + \frac{50}{27}\eta^3 - \frac{\eta^4}{6} - \frac{4}{9}\eta^5 - \frac{2}{27}\eta^6, \tag{9}$$

the new rational verifier recomputes all four interval images and checks

$$I_{\mathrm{shell},j} + [0, 8\ell] F_j(I_{\mathrm{outer}}) \;\subset\; \operatorname{int}(I_{\mathrm{outer},j}).$$

All eight component margins are positive; also $131(8\ell) < 9$, the retained proper endpoint margin. The usual continuation/first-exit argument therefore establishes (7) on the declared static collar. This is interval arithmetic on the ODE, not numerical integration or a new sampled background orbit.

Writing $h = U_1/p$, direct substitution of (8) gives

$$\begin{aligned} C &= \rho(2h - 5a), \\ C_z &= \rho^2(2h^2 - 10ah + 5a^2 - 2U_2 - 5p^2/3) - 5, \\ \mathcal{W} &= \rho^2\mathcal{Q} - \frac{35}{4}\tanh^2 t, \qquad \mathcal{Q} = 2h^2 - 10ah + \frac{35}{4}a^2 - U_2 - \frac{p^2}{6}. \end{aligned} \tag{10}$$

The polynomial and (7) give exact strict guards

$$\frac{7}{4} < h < \frac{7}{3}, \qquad \frac{31}{10} < U_2 < \frac{25}{8}, \qquad \frac{1}{3} < \mathcal{Q} < 6. \tag{11}$$

For example, expanding $U_1, U_2$ at $\eta = 0$ and bounding each monomial gives the rational comparisons

$$\mathcal{Q} > \frac{1933333317}{5000000000} > \frac{1}{3}, \qquad \mathcal{Q} < \frac{2680267}{450000} < 6.$$

Since $|\tanh t| \le |t|$ for real $t$, (10) yields

$$\begin{aligned} -\ell \le t \le 0, \quad 0 \le z \le 8\ell : \qquad & 5500 < \mathcal{W} < 103000, \\ & 0 < \mathcal{V} < \mu, \qquad \mu = \frac{103}{1000000000}. \end{aligned} \tag{12}$$

Indeed $129^2/3 - 35\ell^2/4 > 5500$ and $131^2\,6 + 35\ell^2/4 < 103000$. The explicit time dependence uses the same positive-depth static branch. No future five-mean operator cap, positive-time evolved error, or physical negative-$z$ background is transferred to the past.

## 3 The backward Cauchy solution before shell contact

Fix $0 < \tau < 1$ and put $s = -x$. The required support cone is

$$1 - s \le r \le 2 + s, \qquad 0 \le s \le \tau. \tag{13}$$

It lies in $(0, 3)$, inside the verified collar. Extend $\mathcal{V}$ smoothly to an auxiliary whole-line coefficient, unchanged near this compact cone and with absolute bound at most $\mu$. The solution constructed below is supported in (13); thus it is independent of this auxiliary extension. No physical background outside the positive collar is introduced.

Write $\widehat{U}(s, r) = U(-s, r)$. The free backward wave and the actual equation are

$$\widehat{U}_f(s, r) = \frac{Q(r - s) + Q(r + s)}{2} - \frac{\ell}{2}\int_{r-s}^{r+s} e^{-\ell y} Q(y)\,\mathrm{d}y, \tag{14}$$

$$\widehat{U}(s, r) = \widehat{U}_f(s, r) - \frac{1}{2}\int_0^s \int_{r-s+\xi}^{r+s-\xi} \mathcal{V}(-\xi, y)\widehat{U}(\xi, y)\,\mathrm{d}y\,\mathrm{d}\xi. \tag{15}$$

The minus sign in (14) comes from the actual nonzero present velocity in (6). On continuous functions with support (13), the integral operator $K$ has norm at most $\beta_\tau = \mu\tau^2/2 < 1$. Hence $\widehat{U} = \sum_{n\ge 0}(-K)^n\widehat{U}_f$ converges uniformly, constructs the solution, and proves uniqueness and continuous dependence in this class. Its tail after order $N$ is bounded by $\beta_\tau^{N+1}(1 + \tau\ell)/(1 - \beta_\tau)$. Differentiated

Volterra equations give smoothness for the smooth compact data and coefficient. Higher coefficient derivatives are bounded on each fixed compact cone; they are not needed in the following norm bounds.

Here $\|\cdot\|_2$ integrates in $r$ over the auxiliary whole line, equivalently over the positive collar on this supported solution. Translations and the $L^1$ norm of the interval convolution kernel yield

$$\begin{aligned} \sup_{0\le s\le\tau}\|\widehat{U}_f(s)\|_2 \le (1+\tau\ell)\|Q\|_2, \quad \sup\|\widehat{U}(s)\|_2 \le B_\tau\|Q\|_2, \\ \sup|\widehat{U}| \le B_\tau, \qquad\qquad B_\tau = \frac{1+\tau\ell}{1-\mu\tau^2/2}. \end{aligned} \tag{16}$$

For the pointwise bound use $0 \le Q \le 1$. All these constants are bounded uniformly for $\tau < 1$ by

$$\beta = \frac{\mu}{2} = \frac{103}{2000000000} < 1, \qquad B = \frac{1+\ell}{1-\mu/2} = \frac{2000002000}{1999999897} < 1.000002. \tag{17}$$

Uniqueness identifies the solutions on overlapping closed slabs. Their union gives the actual local linear solution on $-1 < x \le 0$. Its support has a strictly positive shell gap at every such time. For example the quarter-slab support lies in $[3/4, 9/4]$ and the half-slab support in $[1/2, 5/2]$. The limiting front reaches $r = 0$ only at $x = -1$. No endpoint interaction or earlier-past assertion is made.

## 4 Persistence of both normalized principal channels

Let $D_Q = \|Q_r\|_2 > 0$. The two-endpoint Cauchy inequality on the unit interval gives $\|Q\|_2 \le D_Q/2$. Since $Q(3/2) = 1$ and $Q(1) = Q(2) = 0$, applying Cauchy separately to the two half intervals also gives $D_Q^2 \ge 4$. These analytic bump facts require no new bump sampling. The initial norms of $J_\pm$ lie between $(1-\ell/2)D_Q$ and $(1+\ell/2)D_Q$. The exact characteristic equations are

$$(\partial_x - \partial_r)J_+ = -\mathcal{V}U, \qquad (\partial_x + \partial_r)J_- = -\mathcal{V}U. \tag{18}$$

Translation preserves $L^2$. Their past integrals and (16)–(17) therefore imply

$$\|J_\pm(x) - \text{translated } J_\pm(0)\|_2 \le \mu|x|B\|Q\|_2 \le \frac{\mu B}{2}D_Q.$$

Consequently, throughout $-1 < x \le 0$,

$$(1-\delta)D_Q \le \|J_\pm(x)\|_2 \le (1+\delta)D_Q, \qquad \delta = \frac{\ell}{2} + \frac{\mu B}{2} = \frac{2206000103}{3999999794000000} < 10^{-6}. \tag{19}$$

In particular neither channel vanishes on an entire such Cauchy slice. For $E_\pm(x) = \|J_\pm(x)\|_2^2/4$, monotonicity of a ratio gives

$$\frac{(1-\delta)^2}{(1-\delta)^2+(1+\delta)^2} \le \frac{E_\pm}{E_+ + E_-} \le \frac{(1+\delta)^2}{(1-\delta)^2+(1+\delta)^2}. \tag{20}$$

The exact rational guards yield the simple strict display

$$0.499999 < \frac{E_\pm(x)}{E_+(x)+E_-(x)} < 0.500001, \qquad -1 < x \le 0. \tag{21}$$

Thus data with either $J_- = 0$ or $J_+ = 0$ on an entire slice in this interval cannot evolve into the registered data under the stated equation. The unscaled $t, z$ weights differ by a common factor $1/\ell$, so their shares agree. A spatially varying conjugation does not preserve these weights. They are normalized-master principal quadratic diagnostics, not physical canonical or Noether energies, radiation fluxes, or conserved quantities in the variable-potential problem.

## 5 Two compatible past characteristic traces

Use null coordinates $u = x + r$, $v = x - r$ and the half-diamond

$$\mathcal{D} = \{1 \le u \le 2,\ -2 \le v \le -1,\ u + v \le 0\}. \tag{22}$$

Its present base is the entire packet interval $x = 0$, $1 \le r \le 2$. Its past apex $A = (-1/2, 3/2)$ lies inside the positive collar, not on the shell. Its past faces are $u = 1$ and $v = -2$. They are not the shell-corner cut of the separate second-order affine-null preparation discussion [3]. Restrict the solution from Section 3 to define

$$\begin{aligned} f(v) &= U((1+v)/2, (1-v)/2), & -2 \le v \le -1, \\ g(u) &= U((u-2)/2, (u+2)/2), & 1 \le u \le 2, \\ c &= U(-1/2, 3/2) = f(-2) = g(1). \end{aligned} \tag{23}$$

The common corner identity is exact; $c$ is not an independently chosen zero. It is scalar trace compatibility for this local wave problem, not full nonlinear null or shell compatibility.

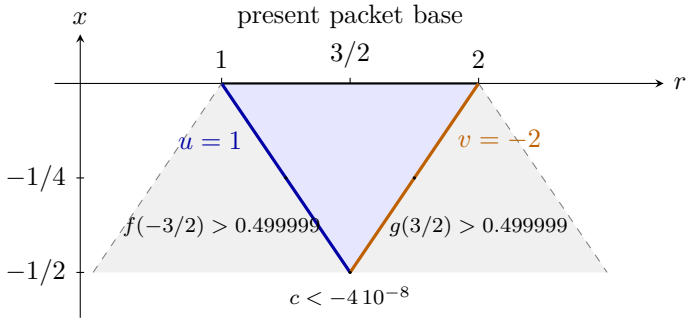



Figure 1: The actual characteristic half-diamond in $(r, x)$ coordinates. The gray trapezoid indicates the half-time Cauchy support cone. The two face midpoints are positive and their shared interior apex is negative. The shell $r = 0$ lies outside the diagram and is not contacted.

Put $U_f(x, r) = \widehat{U}_f(-x, r)$. For the half slab let

$$\beta_{1/2} = \frac{\mu}{8}, \qquad B_{1/2} = \frac{1 + \ell/2}{1 - \mu/8}, \qquad \epsilon_* = \beta_{1/2} B_{1/2} = \frac{206000103}{15999999794000000}. \tag{24}$$

Equation (15) gives $|U - U_f| \le \epsilon_*$. The free endpoint arguments at $(-1/4, 5/4)$ are $1, 3/2$, and those at $(-1/4, 7/4)$ are $3/2, 2$. Hence

$$f(-3/2),\ g(3/2) \ge \frac{1}{2} - \frac{\ell}{4} - \epsilon_* > 0.499999. \tag{25}$$

At $A$ the two free displacement terms vanish, so

$$U_f(A) = -\frac{\ell}{2} \int_1^2 e^{-\ell r} Q(r)\, \mathrm{d}r.$$

On $[5/4, 7/4]$, $Q \ge e^{-4/3} > 1/4$. For instance the positive Taylor sum and geometric tail give

$$e^{4/3} \le 1 + \frac{4}{3} + \frac{(4/3)^2}{2} + \frac{(4/3)^3/6}{1 - (4/3)/4} = \frac{103}{27} < 4.$$

Together with $e^{-\ell r} \geq 1 - 2\ell$, this proves

$$c \leq -\frac{\ell(1-2\ell)}{16} + \epsilon_* < -\frac{1}{25000000} = -4\,10^{-8}. \tag{26}$$

The negative corner depends on the registered nonzero velocity. It would not follow from the same proof for zero initial velocity.

Each continuous trace therefore has an internal zero between its apex and midpoint and is nonconstant. Each has a nonzero tangential derivative somewhere on that half-face. Precisely,

$$2f_v = J_- \quad (u = 1), \qquad 2g_u = J_+ \quad (v = -2). \tag{27}$$

These are derivatives along the faces, not amplitudes transported in the same direction as those faces. Thus prescribing either face trace to be the constant $c$, or the corresponding whole-face tangential principal derivative to vanish, cannot produce this packet on $\mathcal{D}$. The zeros and sign changes survive positive smooth master conjugation and orientation-preserving face reparametrization. Numerical channel weights and derivative sizes do not remain unchanged under those operations.

In formulas with null arguments, $U(u, v)$ denotes $U((u+v)/2, (u-v)/2)$. The wave equation is $4U_{uv} + \mathcal{V}U = 0$. Its Goursat formula is

$$U(u,v) = f(v) + g(u) - c - \frac{1}{4}\int_1^u \int_{-2}^v \mathcal{V}((a+b)/2, (a-b)/2)U(a,b)\,\mathrm{d}b\,\mathrm{d}a. \tag{28}$$

Every integration rectangle stays inside $\mathcal{D}$, and $(u-1) + (v+2) \leq 1$ bounds its area by $1/4$. The operator norm is therefore at most

$$\frac{\mu}{16} = \frac{103}{16000000000} < 1. \tag{29}$$

The convergent Volterra solution is unique and smooth for these compatible smooth traces. It equals the previously constructed Cauchy solution and reproduces both its present packet and its velocity. This local two-face argument does not infer global retarded uniqueness from one unrestricted null cut; such assertions need causal and trace-space hypotheses [1].

## 6 Coupled reconstruction, physical scope, and verification

The master is not an unrelated test scalar. For the fixed sphere harmonic $Y = n_1^2 - n_2^2$, the retained differential reconstruction is

$$\begin{aligned} P &= \mathsf{V}Y, \qquad \Psi = P/\rho^2, \qquad h_{yy} = -4\Psi, \qquad h_{y\mu} = 0, \\ h_{\mu\nu} &= 2\Psi\, g_{0,\mu\nu}, \qquad \delta\phi = -3P_y/(\rho^2 p). \end{aligned} \tag{30}$$

Here $\mu, \nu$ range over the four de Sitter directions in (3). Under the retained identities, (4) implies the full first linearized Einstein–scalar equations and constraints. The same static ODE and smooth hyperbolic functions apply on the verified negative-time domain. For every compact past slab with $\tau < 1$, all first-field jets vanish on an open shell collar by (13). The physical first linear junction rows are then background there. Identification with the fixed common-harmonic representative still uses the full canonical one-jet, including the mixed metric velocity, and the retained gauge/uniqueness premises. No future gauge-error enclosure is used as a past error bound.

The characteristic data constructed here are restrictions of a selected solution. They are not an independently selected physical state, arbitrary incoming data, or an onto map to all prescribed

histories. Their common negative corner prevents interpreting them as two independently flat, zero-corner bumps. A full nonlinear past problem must still supply and propagate the other null constraints, the scalar equation, physical shell conditions, and compatible free state data. The constant-umbilic vacuum boundary theorem of Fournodavlos–Smulevici [2] is a useful comparison, but its homogeneous boundary momentum-constraint mechanism does not by itself handle a variable factor $\sigma(\phi)/6$ and scalar flux in five dimensions.

A conditional exact-family interpretation would additionally require a time-reversed local development of an exact constraint curve and the corresponding gauge identification. It is not proved by the linear Volterra calculation. Higher-order constraint corrections can have a larger support annulus than the first packet, so their contact-free interval must not be inferred from (13). No uniform effective nonlinear parameter window, second constrained preparation, source or injected-energy law is obtained. Earlier incoming states, potential scattering, or earlier shell interaction remain possible. The result supplies no matter or radiation production, absolute proper shell-observable sign, measured scale, or cosmological origin.

The accompanying exact verifier authenticates five local input copies: two numeric enclosures and three selected analytic/state snapshots. It recomputes four rational ODE images per successful process and the potential, Volterra, channel, and trace guards. Normal and optimized Python produce identical canonical reports. There are no project imports, legacy checker calls, background/source/bump oracle queries, or numerical PDE solves. Analytic reconstruction and branch premises remain explicit. Independent coefficient and characteristic reviews, a displayed-bounds audit, two fresh deterministic three-pass PDF builds, and all-page visual review accompany this working manuscript.

## References


[1] C. Bär and R. Tagne Wafo, *Initial value problems for wave equations on manifolds*, Math. Phys. Anal. Geom. **18**, article 7 (2015), arXiv:1408.4995v2. Theorem 8, Corollary 10, and Theorem 23 distinguish finite-slab control from characteristic existence with causal hypotheses.

[2] G. Fournodavlos and J. Smulevici, *The initial boundary value problem in General Relativity: the umbilic case*, arXiv:2104.08851v1. Theorem 1 and Lemma 1 concern a constant-umbilic four-dimensional vacuum boundary; Appendix A treats scalar Robin estimates.

[3] R. Maldonado, *Incoming preparation constraints and quadratic source budgets for scalar shells*, working manuscript (4 November 2026), dated research capsule. The present capsule retains the selected source snapshots and records the new past-domain calculation separately.